

EDEXCEL INTERNATIONAL A LEVEL

# WMA11/01 May/June 2024 Worked Solutions

May/June 2024 paper rebuilt with the worked solution after each question.

**11**

paper questions

**WMA11**

Pure 1

**Single-  
paper  
solutions**standalone IAL  
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - [eliteigcse.com](https://eliteigcse.com)**P1**

**PAST PAPER**

# **WMA11/01 May/June 2024**

**May/June 2024 | 11 questions | 75 marks**

**11**

questions

**75**

marks

**Answers**

worked solution  
after each  
question

## Question 1

## Integration

1. Find

$$\int \left( 10x^4 - \frac{3}{2x^3} - 7 \right) dx$$

giving each term in simplest form.

(3)

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(Total for Question 1 is 3 marks)

# Worked Solution - Question 1

**1. Rewrite the fraction**

$$10x^4 - \frac{3}{2x^3} - 7 = 10x^4 - \frac{3}{2}x^{-3} - 7.$$

**2. Integrate term by term**

$$\int 10x^4 dx = 2x^5, \int -\frac{3}{2}x^{-3} dx = \frac{3}{4}x^{-2}, \text{ and } \int -7 dx = -7x.$$

**3. Write in simplest form**

$$2x^5 + \frac{3}{4x^2} - 7x + c.$$

**Final answer**

$$2x^5 + \frac{3}{4x^2} - 7x + c.$$

## Question 2

## Indices &amp; Surds

2. (i) Given that  $m = 2^n$ , express each of the following in simplest form in terms of  $m$ .

(a)  $2^{n+3}$

(1)

(b)  $16^{3n}$

(2)

- (ii) **In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

Solve the equation

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

giving your answer in the form  $p + q\sqrt{3}$  where  $p$  and  $q$  are integers.

(3)

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(Total for Question 2 is 6 marks)

## Worked Solution - Question 2

**1. Use m equals 2 to the n**

$$2^{n+3} = 2^n \times 2^3 = 8m.$$

**2. Rewrite 16 as a power of 2**

$$16^{3n} = (2^4)^{3n} = 2^{12n} = (2^n)^{12} = m^{12}.$$

**3. Collect the x terms**

$$x\sqrt{3} - 3 = x + \sqrt{3} \text{ gives } x(\sqrt{3} - 1) = 3 + \sqrt{3}.$$

**4. Rationalise**

$$x = \frac{3 + \sqrt{3}}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = \frac{(3 + \sqrt{3})(\sqrt{3} + 1)}{2}.$$

**5. Simplify**

$$x = \frac{4\sqrt{3} + 6}{2} = 3 + 2\sqrt{3}.$$

**Final answer**

$$(i)(a) 8m. \quad (i)(b) m^{12}. \quad (ii) x = 3 + 2\sqrt{3}.$$

## Question 3

## Transformations

3.

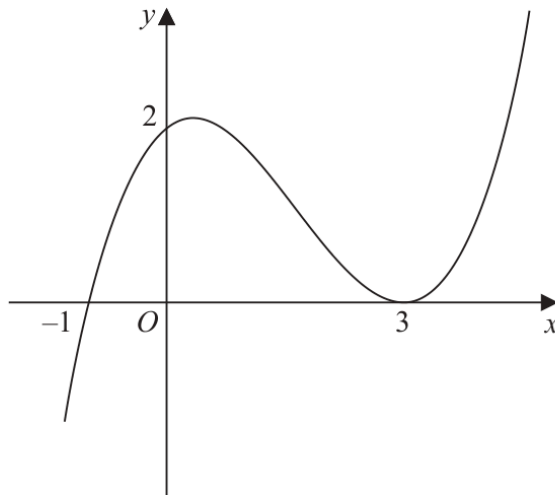


Figure 1

Figure 1 shows a sketch of the curve with equation  $y = f(x)$ .

The curve passes through the points  $(-1, 0)$  and  $(0, 2)$  and touches the  $x$ -axis at the point  $(3, 0)$ .

On separate diagrams, sketch the curve with equation

(a)  $y = f(x + 3)$  (3)

(b)  $y = f(-3x)$  (3)

On each diagram, show clearly the coordinates of all the points where the curve cuts or touches the coordinate axes.

(Total for Question 3 is 6 marks)

# Worked Solution - Question 3

**1. Read the original key points**

The original curve passes through  $(-1, 0)$  and  $(0, 2)$ , and touches the x-axis at  $(3, 0)$ .

**2. Transform  $y$  equals  $f$  of  $x$  plus 3**

$y = f(x + 3)$  is a translation 3 units left. So  $(-1, 0)$  moves to  $(-4, 0)$ ,  $(0, 2)$  moves to  $(-3, 2)$ , and  $(3, 0)$  moves to  $(0, 0)$ .

**3. State the axis intercepts for part a**

The curve cuts the x-axis at  $(-4, 0)$  and touches it at  $(0, 0)$ . The y-intercept is also  $(0, 0)$ .

**4. Transform  $y$  equals  $f$  of negative  $3x$** 

For  $y = f(-3x)$ , a point with original x-coordinate  $X$  moves to  $x = -\frac{X}{3}$ .

**5. State the axis intercepts for part b**

$(-1, 0)$  moves to  $(\frac{1}{3}, 0)$ ,  $(3, 0)$  moves to  $(-1, 0)$ , and  $(0, 2)$  stays at  $(0, 2)$ .

**Final answer**

(a) intercepts  $(-4, 0)$  and  $(0, 0)$ , with  $(0, 0)$  a touch point.

(b) intercepts  $(-1, 0)$ ,  $(\frac{1}{3}, 0)$  and  $(0, 2)$ , with  $(-1, 0)$  a touch point.

## Question 4

## Quadratics

4. The curve  $C_1$  has equation

$$y = x^2 + kx - 9$$

and the curve  $C_2$  has equation

$$y = -3x^2 - 5x + k$$

where  $k$  is a constant.

Given that  $C_1$  and  $C_2$  meet at a single point  $P$

- (a) show that

$$k^2 + 26k + 169 = 0 \quad (3)$$

- (b) Hence find the coordinates of  $P$

(3)

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(Total for Question 4 is 6 marks)

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## Worked Solution - Question 4

**1. Equate the curves**

$$x^2 + kx - 9 = -3x^2 - 5x + k \text{ gives } 4x^2 + (k + 5)x - (k + 9) = 0.$$

**2. Use one intersection only**

For a single point of intersection, the discriminant is zero.

**3. Form the condition on  $k$** 

$$(k + 5)^2 - 4(4)(-(k + 9)) = 0, \text{ so } k^2 + 26k + 169 = 0.$$

**4. Find  $k$** 

$$k^2 + 26k + 169 = (k + 13)^2, \text{ so } k = -13.$$

**5. Find  $x$** 

Substituting  $k = -13$  gives  $4x^2 - 8x + 4 = 0$ , so  $4(x - 1)^2 = 0$  and  $x = 1$ .

**6. Find  $y$** 

$$y = x^2 + kx - 9 = 1 - 13 - 9 = -21, \text{ so } P = (1, -21).$$

**Final answer**

$$P = (1, -21).$$

## Question 5

Radians

5.

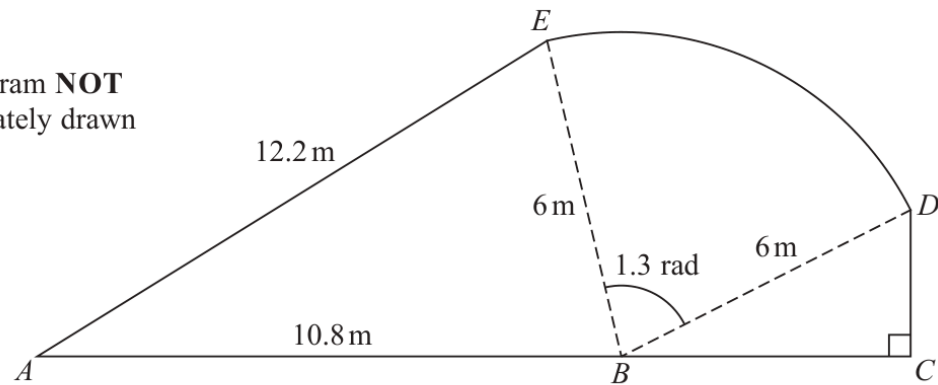
Diagram NOT  
accurately drawn

Figure 2

Figure 2 shows the plan view of a garden.

The shape of the garden  $ABCDEA$  consists of a triangle  $ABE$  and a right-angled triangle  $BCD$  joined to a sector  $BDE$  of a circle with radius 6 m and centre  $B$ .

The points  $A$ ,  $B$  and  $C$  lie on a straight line with  $AB = 10.8$  m

Angle  $BCD = \frac{\pi}{2}$  radians, angle  $EBD = 1.3$  radians and  $AE = 12.2$  m

- (a) Find the area of the sector  $BDE$ , giving your answer in  $\text{m}^2$  (2)
- (b) Find the size of angle  $ABE$ , giving your answer in radians to 2 decimal places. (2)
- (c) Find the area of the garden, giving your answer in  $\text{m}^2$  to 3 significant figures. (3)

(Total for Question 5 is 7 marks)

## Worked Solution - Question 5

**1. Find the sector area**

$$\text{Area of sector } BDE = \frac{1}{2}(6^2)(1.3) = 23.4 \text{ m}^2.$$

**2. Use the cosine rule in triangle ABE**

$$12.2^2 = 10.8^2 + 6^2 - 2(10.8)(6) \cos \angle ABE.$$

**3. Find angle ABE**

$$\cos \angle ABE = \frac{10.8^2 + 6^2 - 12.2^2}{2(10.8)(6)}, \text{ so } \angle ABE \approx 1.54 \text{ radians.}$$

**4. Find the area of triangle ABE**

$$\text{Area } ABE = \frac{1}{2}(10.8)(6) \sin(1.54147\dots) \approx 32.386.$$

**5. Find angle DBC**

$$\angle DBC = \pi - 1.3 - \angle ABE \approx 0.30012 \text{ radians.}$$

**6. Find the area of triangle BCD**

$$\text{Area } BCD = \frac{1}{2}(6 \cos \angle DBC)(6 \sin \angle DBC) \approx 5.084.$$

**7. Add the three areas**

$$23.4 + 32.386 + 5.084 \approx 60.9 \text{ m}^2 \text{ to 3 significant figures.}$$

**Final answer**

(a)  $23.4 \text{ m}^2$ .

(b) 1.54 radians.

(c)  $60.9 \text{ m}^2$ .

## Question 6

## Inequalities

6.

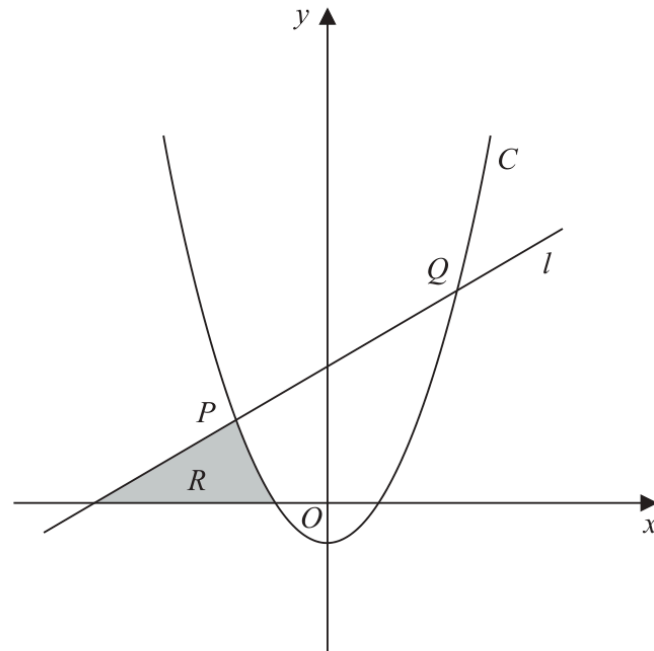


Figure 3

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows

- the line  $l$  with equation  $y - 5x = 75$
- the curve  $C$  with equation  $y = 2x^2 + x - 21$

The line  $l$  intersects the curve  $C$  at the points  $P$  and  $Q$ , as shown in Figure 3.

- (a) Find, using algebra, the coordinates of  $P$  and the coordinates of  $Q$ .

(4)

The region  $R$ , shown shaded in Figure 3, is bounded by  $C$ ,  $l$  and the  $x$ -axis.

- (b) Use inequalities to define the region  $R$ .

(3)

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(Total for Question 6 is 7 marks)

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## Worked Solution - Question 6

**1. Write the line as  $y$  equals**

$$y - 5x = 75, \text{ so } y = 5x + 75.$$

**2. Equate line and curve**

$$5x + 75 = 2x^2 + x - 21, \text{ so } 2x^2 - 4x - 96 = 0.$$

**3. Solve the quadratic**

$$x^2 - 2x - 48 = 0 = (x - 8)(x + 6), \text{ so } x = -6 \text{ or } x = 8.$$

**4. Find the  $y$ -coordinates**

Using  $y = 5x + 75$ ,  $x = -6$  gives  $y = 45$  and  $x = 8$  gives  $y = 115$ .

**5. State  $P$  and  $Q$** 

From the diagram,  $P = (-6, 45)$  and  $Q = (8, 115)$ .

**6. Use the  $x$ -axis and left branch**

The shaded region is above the  $x$ -axis and on the left part of the diagram, so  $y \geq 0$  and  $x \leq 0$ .

**7. Use the line and curve as upper boundaries**

The region is below both the line and the curve, so  $y \leq 5x + 75$  and  $y \leq 2x^2 + x - 21$ .

**Final answer**

(a)  $P = (-6, 45)$ ,  $Q = (8, 115)$ .

(b)  $y \geq 0$ ,  $y \leq 5x + 75$ ,  $y \leq 2x^2 + x - 21$ ,  $x \leq 0$ .

## Question 7

## Differentiation

7. The curve  $C$  has equation  $y = f(x)$  where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and  $k$  is a constant.

- (a) Find, in simplest form,

(i)  $f'(x)$

(ii)  $f''(x)$

(3)

The curve with equation  $y = f'(x)$  intersects the curve with equation  $y = f''(x)$  at the points  $A$  and  $B$ .

Given that the  $x$  coordinate of  $A$  is 5

- (b) find the value of  $k$ .

(2)

- (c) Hence find the coordinates of  $B$ .

(3)

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(Total for Question 7 is 8 marks)

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## Worked Solution - Question 7

**1. Differentiate once**

$$f'(x) = 6x^2 - 2kx + 14.$$

**2. Differentiate again**

$$f''(x) = 12x - 2k.$$

**3. Use the intersection at A**

At an intersection of  $y = f'(x)$  and  $y = f''(x)$ ,  $f'(x) = f''(x)$ . Given  $x_A = 5$ , substitute  $x = 5$ .

**4. Find k**

$$6(5)^2 - 2k(5) + 14 = 12(5) - 2k, \text{ so } 164 - 10k = 60 - 2k \text{ and } k = 13.$$

**5. Find the other x-coordinate**

With  $k = 13$ ,  $6x^2 - 26x + 14 = 12x - 26$ , so  $6x^2 - 38x + 40 = 0$ . Hence  $(3x - 4)(x - 5) = 0$ .

**6. Use the point B**

The other point has  $x = \frac{4}{3}$ . Then  $y = f''(\frac{4}{3}) = 12(\frac{4}{3}) - 26 = -10$ .

**Final answer**

$$(a)(i) f'(x) = 6x^2 - 2kx + 14. \quad (a)(ii) f''(x) = 12x - 2k.$$

$$(b) k = 13.$$

$$(c) B = (\frac{4}{3}, -10).$$

## Question 8

## Graphs of Functions

8. The curve  $C_1$  has equation

$$y = x(4 - x^2)$$

- (a) Sketch the graph of  $C_1$  showing the coordinates of any points of intersection with the coordinate axes.

(3)

The curve  $C_2$  has equation  $y = \frac{A}{x}$  where  $A$  is a constant.

- (b) Show that the  $x$  coordinates of the points of intersection of  $C_1$  and  $C_2$  satisfy the equation

$$x^4 - 4x^2 + A = 0$$

(1)

- (c) Hence find the range of possible values of  $A$  for which  $C_1$  meets  $C_2$  at 4 distinct points.

(3)

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(Total for Question 8 is 7 marks)

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## Worked Solution - Question 8

**1. Find the intercepts of C1**

$y = x(4 - x^2)$ , so  $y = 0$  when  $x = 0$  or  $x = \pm 2$ . The curve crosses the axes at  $(-2, 0)$ ,  $(0, 0)$  and  $(2, 0)$ .

**2. Use the cubic shape**

$y = 4x - x^3$  has negative leading cubic term, so the left tail rises and the right tail falls.

**3. Equate C1 and C2**

$$x(4 - x^2) = \frac{A}{x}.$$

**4. Multiply by x**

$x^2(4 - x^2) = A$ , so  $4x^2 - x^4 = A$  and therefore  $x^4 - 4x^2 + A = 0$ .

**5. Let u equal x squared**

Let  $u = x^2$ . Then  $u^2 - 4u + A = 0$ .

**6. Require two positive u-values**

For 4 distinct x-values, the quadratic in  $u$  must have two distinct positive roots.

**7. Find the range of A**

The discriminant gives  $16 - 4A > 0$ , so  $A < 4$ . The product of the two roots is  $A$ , so  $A > 0$ . Therefore  $0 < A < 4$ .

**Final answer**

(a) x-intercepts  $(-2, 0)$ ,  $(0, 0)$ ,  $(2, 0)$ .

(b)  $x^4 - 4x^2 + A = 0$ .

(c)  $0$

## Question 9

Straight Line

9. Given that

- the point  $A$  has coordinates  $(4, 2)$
- the point  $B$  has coordinates  $(15, 7)$
- the line  $l_1$  passes through  $A$  and  $B$

(a) find an equation for  $l_1$ , giving your answer in the form  $px + qy + r = 0$  where  $p$ ,  $q$  and  $r$  are integers to be found.

(3)

The line  $l_2$  passes through  $A$  and is parallel to the  $x$ -axis.

The point  $C$  lies on  $l_2$  so that the length of  $BC$  is  $5\sqrt{5}$

(b) Find both possible pairs of coordinates of the point  $C$ .

(4)

(c) Hence find the minimum possible area of triangle  $ABC$ .

(2)

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(Total for Question 9 is 9 marks)

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## Worked Solution - Question 9

**1. Find the gradient of  $l_1$** 

$$m = \frac{7 - 2}{15 - 4} = \frac{5}{11}.$$

**2. Use point A**

$$y - 2 = \frac{5}{11}(x - 4).$$

**3. Rearrange**

$$11y - 22 = 5x - 20, \text{ so } 5x - 11y + 2 = 0.$$

**4. Use the horizontal line  $l_2$** 

Since  $l_2$  passes through  $A(4, 2)$  and is parallel to the x-axis, point  $C$  has coordinates  $(c, 2)$ .

**5. Use BC equals 5 root 5**

$$(c - 15)^2 + (2 - 7)^2 = (5\sqrt{5})^2.$$

**6. Solve for c**

$$(c - 15)^2 + 25 = 125, \text{ so } (c - 15)^2 = 100 \text{ and } c = 5 \text{ or } 25.$$

**7. Find the smaller area**

The height from  $B$  to  $y = 2$  is 5. For  $C = (5, 2)$ ,  $AC = 1$ , so the minimum area is  $\frac{1}{2}(1)(5) = \frac{5}{2}$ .

**Final answer**

(a)  $5x - 11y + 2 = 0$ .

(b)  $C = (5, 2)$  or  $(25, 2)$ .

(c)  $\frac{5}{2}$ .

## Question 10

## Integration

10. The curve  $C$  has equation  $y = f(x)$  where  $x > 0$

Given that

- $f'(x) = 6x - \frac{(2x-1)(3x+2)}{2\sqrt{x}}$

- the point  $P(4, 12)$  lies on  $C$

(a) find the equation of the normal to  $C$  at  $P$ , giving your answer in the form  $y = mx + c$  where  $m$  and  $c$  are integers to be found,

(4)

(b) find  $f(x)$ , giving each term in simplest form.

(6)

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(Total for Question 10 is 10 marks)

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## Worked Solution - Question 10

**1. Expand the numerator**

$$(2x - 1)(3x + 2) = 6x^2 + x - 2.$$

**2. Simplify f prime**

$$f'(x) = 6x - \frac{6x^2 + x - 2}{2\sqrt{x}} = 6x - 3x^{3/2} - \frac{1}{2}x^{1/2} + x^{-1/2}.$$

**3. Find the tangent gradient at P**

$$f'(4) = 24 - 3(8) - \frac{1}{2}(2) + \frac{1}{2} = -\frac{1}{2}.$$

**4. Use the normal gradient**

The normal gradient is the negative reciprocal of  $-\frac{1}{2}$ , so it is 2.

**5. Find the normal equation**

Through  $P(4, 12)$ ,  $y - 12 = 2(x - 4)$ , hence  $y = 2x + 4$ .

**6. Integrate f prime**

$$f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + c.$$

**7. Use P to find c**

$$12 = 3(4)^2 - \frac{6}{5}(4^{5/2}) - \frac{1}{3}(4^{3/2}) + 2(4^{1/2}) + c.$$

**8. Simplify c**

$$12 = 48 - \frac{192}{5} - \frac{8}{3} + 4 + c = \frac{164}{15} + c, \text{ so } c = \frac{16}{15}.$$

**Final answer**

(a)  $y = 2x + 4$ .

(b)  $f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + \frac{16}{15}$ .

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6 marks

## Question 11

Trigonometric Functions

11.

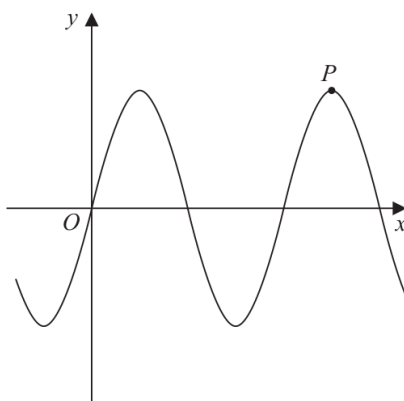


Figure 4

Figure 4 shows a sketch of part of the curve  $C_1$  with equation

$$y = 12 \sin x$$

where  $x$  is measured in radians.

The point  $P$  shown in Figure 4 is a maximum point on  $C_1$

(a) Find the coordinates of  $P$ .

(2)

The curve  $C_2$  has equation

$$y = 12 \sin x + k$$

where  $k$  is a constant.

Given that the **maximum** value of  $y$  on  $C_2$  is 3

(b) find the coordinates of the **minimum** point on  $C_2$  which has the **smallest** positive  $x$  coordinate.

(2)

The curve  $C_3$  has equation

$$y = 12 \sin(x + B)$$

where  $B$  is a positive constant.

Given that  $\left(\frac{\pi}{4}, A\right)$ , where  $A$  is a constant, is the **minimum** point on  $C_3$  which has the **smallest** positive  $x$  coordinate,

(c) find

(i) the value of  $A$ ,

(ii) the smallest possible value of  $B$ .

(2)

(Total for Question 11 is 6 marks)

TOTAL FOR PAPER IS 75 MARKS

# Worked Solution - Question 11

**1. Use maxima of sine**

For  $y = 12 \sin x$ , maxima occur when  $x = \frac{\pi}{2} + 2\pi n$ , with value 12.

**2. Read point P**

The maximum point  $P$  shown is the second positive maximum, so  $P = (\frac{5\pi}{2}, 12)$ .

**3. Find  $k$  for C2**

For  $y = 12 \sin x + k$ , the maximum value is  $12 + k$ . Given the maximum is 3,  $k = -9$ .

**4. Find the smallest positive minimum on C2**

The minimum value is  $-12 - 9 = -21$ , and it first occurs at  $x = \frac{3\pi}{2}$ . So the point is  $(\frac{3\pi}{2}, -21)$ .

**5. Find  $A$  for C3**

For  $y = 12 \sin(x + B)$ , any minimum has value  $-12$ , so  $A = -12$ .

**6. Find the smallest positive  $B$** 

At the minimum  $x = \frac{\pi}{4}$ , so  $\frac{\pi}{4} + B = \frac{3\pi}{2}$ . Hence  $B = \frac{5\pi}{4}$ .

**Final answer**

(a)  $P = (\frac{5\pi}{2}, 12)$ .

(b)  $(\frac{3\pi}{2}, -21)$ .

(c)(i)  $A = -12$ .

(c)(ii)  $B = \frac{5\pi}{4}$ .